

HW 9

6.6

Suppose that X, Y are jointly continuous with joint probability density function

$$f(x, y) = \begin{cases} xe^{-x(1+y)} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

a

Find the marginal density functions of X and Y .

✓ Answer ✓

$$f_X(x) = \int_0^{\infty} xe^{-x(1+y)} dy = xe^{-x}x^{-1} = e^{-x}$$
$$f_Y(y) = \int_0^{\infty} xe^{-x(1+y)} dx = (y+1)^{-2}$$

b

Calculate the expectation $E[XY]$

✓ Answer

$$E[XY] = E[X E[Y|X]] = E\left[\frac{X}{X}\right] = 1$$

c

Calculate the expectation $E\left[\frac{X}{1+Y}\right]$

✓ Answer

$$\int_0^{\infty} \frac{1}{1+y} \int_0^{\infty} x^2 e^{-x(1+y)} dx dy = \int_0^{\infty} 2(1+y)^{-4} dy = \frac{2}{3}$$

6.7

Consider the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Suppose that (X, Y) is a uniformly chosen random point from this triangle.

a

Find the marginal density functions X and Y .

✓ Answer

$$f_X(x) = 2 - 2x \quad 0 \leq x \leq 1$$

$$f_Y(y) = 2 - 2y \quad 0 \leq y \leq 1$$

b

Calculate the expectations $E[X]$ and $E[Y]$

✓ Answer

$$E[X] = \frac{2}{3}$$

$$E[Y] = \frac{1}{3}$$

c

Calculate the expectation $E[XY]$

✓ Answer

$$\begin{aligned} 2 \int_0^1 x \int_0^{1-x} y \, dy \, dx &= \int_0^1 x - 2x^2 + x^3 \, dx \\ &= \frac{1}{2} - \frac{2}{3} + \frac{1}{4} = \frac{1}{12} \end{aligned}$$

6.11

Let X have density function

$$f_X(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{else} \end{cases}$$

and let Y be uniformly distributed on the interval $(1, 2)$. Assume X and Y are independent. Give the joint density function of (X, Y) . Calculate the probability $P(Y - X \geq \frac{3}{2})$.

✓ Answer

$$f_{X,Y}(x, y) = \begin{cases} 2x & 0 < x < 1, 1 \leq y \leq 2 \\ 0 & \text{else} \end{cases}$$

$$\int_0^{1/2} \int_{x+\frac{3}{2}}^2 2x \, dy \, dx = \int_0^{1/2} 2x \left(\frac{1}{2} - x \right) \, dx = \frac{1}{8} - \frac{2}{24} = \frac{1}{24}$$

6.18

Suppose that X and Y are integer valued random variables with joint probability mass function given by

$$p_{X,Y}(a,b) = \begin{cases} \frac{1}{4a} & 1 \leq b \leq a \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

a

Show that this is indeed a joint probability mass function

✓ **Answer**

$$\sum_{a=1}^4 \sum_{b=1}^a \frac{1}{4a} = \sum_{a=1}^4 \frac{1}{4} = 1$$

b

Find the marginal probability mass functions of X and Y

✓ **Answer**

$$p_X(x) = \frac{1}{4}$$

$$p_Y(y) = \left[\frac{25}{28}, \frac{13}{48}, \frac{7}{48}, \frac{3}{48} \right]$$

c

Find $P(X = Y + 1)$

✓ **Answer**

$$\frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}$$

7.4

Suppose that X and Y are independent exponential random variables with parameters $\lambda \neq \mu$. Find the density function of $X + Y$

✓ **Answer**

$$\int_0^s \lambda e^{-\lambda x} \mu e^{-\mu(s-x)} dx = \lambda \mu e^{-\mu s} \frac{1 - e^{-(\lambda-\mu)s}}{\lambda - \mu} = \frac{\lambda \mu}{\lambda - \mu} (e^{-\mu s} - e^{-\lambda s})$$

7.6

We deal 5 cards from a standard deck of 52, one by one. What is the probability that the third is a king, and the fifth is the ace of spades?

✓ Answer

$$\frac{4}{52} \cdot \frac{1}{51} = \frac{1}{663}$$

7.16

Let X be a Poisson random variable with parameter λ and Y an independent Bernoulli random variable with parameter p . Find the probability mass function of $X + Y$

✓ Answer

$$S = X + Y$$

$$\text{For } S = 0, P(S = s) = (1 - p)e^{-\lambda}$$

$$\text{Else, } (1 - p) \frac{e^{-\lambda} \lambda^s}{s!} + p \frac{e^{-\lambda} \lambda^{(s-1)}}{(s-1)!} = e^{-\lambda} \lambda^s s! \left(1 - p + \frac{sp}{\lambda}\right)$$

7.22

Let X and Y be two independent normals with the same $N(\mu, \sigma^2)$ distribution, with μ real and $\sigma^2 > 0$. Are there values μ and σ^2 for which $2X$ and $X + Y$ have the same distribution?

✓ Answer

$X + Y$ will always have more variance than $2X$ since we sample twice, instead of doubling our sample repeatedly, however if our initial distribution has no variance, then our outputs should be equal.

If we ignore the $\sigma^2 > 0$ restriction, then μ is anything, $\sigma^2 = 0$

Otherwise, no

7.28

Let X_1, X_2, X_3 be independent $\text{Exp}(\lambda)$ distributed random variables. Find the probability that $P(X_1 < X_2 < X_3)$

✓ Answer

Since all 3 are symmetrical continuous variables, we know all orderings of 3 samples are equally likely, making this ordering have a chance of $\frac{1}{6}$